

Operators in Python

An **operator** is a symbol that performs an operation on variables and values. Python provides many types of operators.

1. Arithmetic Operators

Used for mathematical calculations.

Operator	Description	Example	Output
+	Addition	5 + 3	8
-	Subtraction	10 - 4	6
*	Multiplication	7 * 3	21
/	Division (float)	10 / 4	2.5
//	Floor Division	10 // 4	2
%	Modulus (remainder)	10 % 3	1
**	Exponentiation	2 ** 3	8

```
a, b = 10, 3
print(a + b) # 13
print(a - b) # 7
print(a * b) # 30
print(a / b) # 3.333...
print(a // b) # 3
print(a % b) # 1
print(a ** b) # 1000
```

2. Comparison (Relational) Operators

Used to compare values, returning a boolean (True/False).

Operator	Description	Example	Output
==	Equal to	5 == 5	True
!=	Not equal to	5 != 3	True
>	Greater than	7 > 2	True
<	Less than	7 < 2	False

>=	Greater than or equal to	7 >= 7	True
<=	Less than or equal to	3 <= 5	True

x, y = 10, 20

print(x == y) # False

print(x != y) # True

print(x > y) # False

print(x < y) # True

print(x >= 10) # True

3. Logical Operators

Used to combine conditional statements.

Operator	Description	Example	Output
and	True if both conditions are True	(5 > 2 and 10 > 5)	True
or	True if at least one condition is True	(5 > 10 or 2 < 3)	True
not	Negates the result	not(5 > 2)	False

a, b = True, False

print(a and b) # False

print(a or b) # True

print(not a) # False

4. Assignment Operators

Used to assign values to variables (also supports compound assignment).

Operator	Example	Equivalent To
=	x = 5	Assign 5 to x
+=	x += 3	x = x + 3
-=	x -= 2	x = x - 2
*=	x *= 4	x = x * 4
/=	x /= 2	x = x / 2
//=	x //= 2	x = x // 2
%=	x %= 2	x = x % 2

**=	x **= 3	x = x ** 3
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x = 10

x += 5 # 15

x *= 2 # 30

print(x)

5. Bitwise Operators

Operate on binary representations of numbers.

Operator	Description	Example	Binary Example
&	Bitwise AND	5 & 3	101 & 011 = 001 → 1
	Bitwise OR	5 3	101 011 = 111 → 7
^	Bitwise XOR	5 ^ 3	101 ^ 011 = 110 → 6
~	Bitwise NOT	~5	~0101 = -(5+1) → -6
<<	Left shift	5 << 1	101 << 1 = 1010 → 10
>>	Right shift	5 >> 1	101 >> 1 = 10 → 2

a, b = 5, 3

print(a & b) # 1

print(a | b) # 7

print(a ^ b) # 6

print(~a) # -6

print(a << 2) # 20

print(a >> 1) # 2

6. Membership Operators

Used to check membership in sequences like strings, lists, tuples, etc.

Operator	Description	Example	Output
in	True if value exists	'a' in 'apple'	True
not in	True if value does not exist	'x' not in 'apple'	True

my_list = [1, 2, 3, 4]

print(2 in my_list) # True

```
print(10 not in my_list) # True
```

7. Identity Operators

Used to compare object memory addresses (not values).

Operator	Description	Example	Output
is	True if both refer to same object	x is y	True/False
is not	True if not same object	x is not y	True/False

```
x = [1, 2, 3]
```

```
y = x
```

```
z = [1, 2, 3]
```

```
print(x is y)    # True (same object in memory)
```

```
print(x is z)    # False (different objects, same content)
```

```
print(x == z)    # True (values are equal)
```
